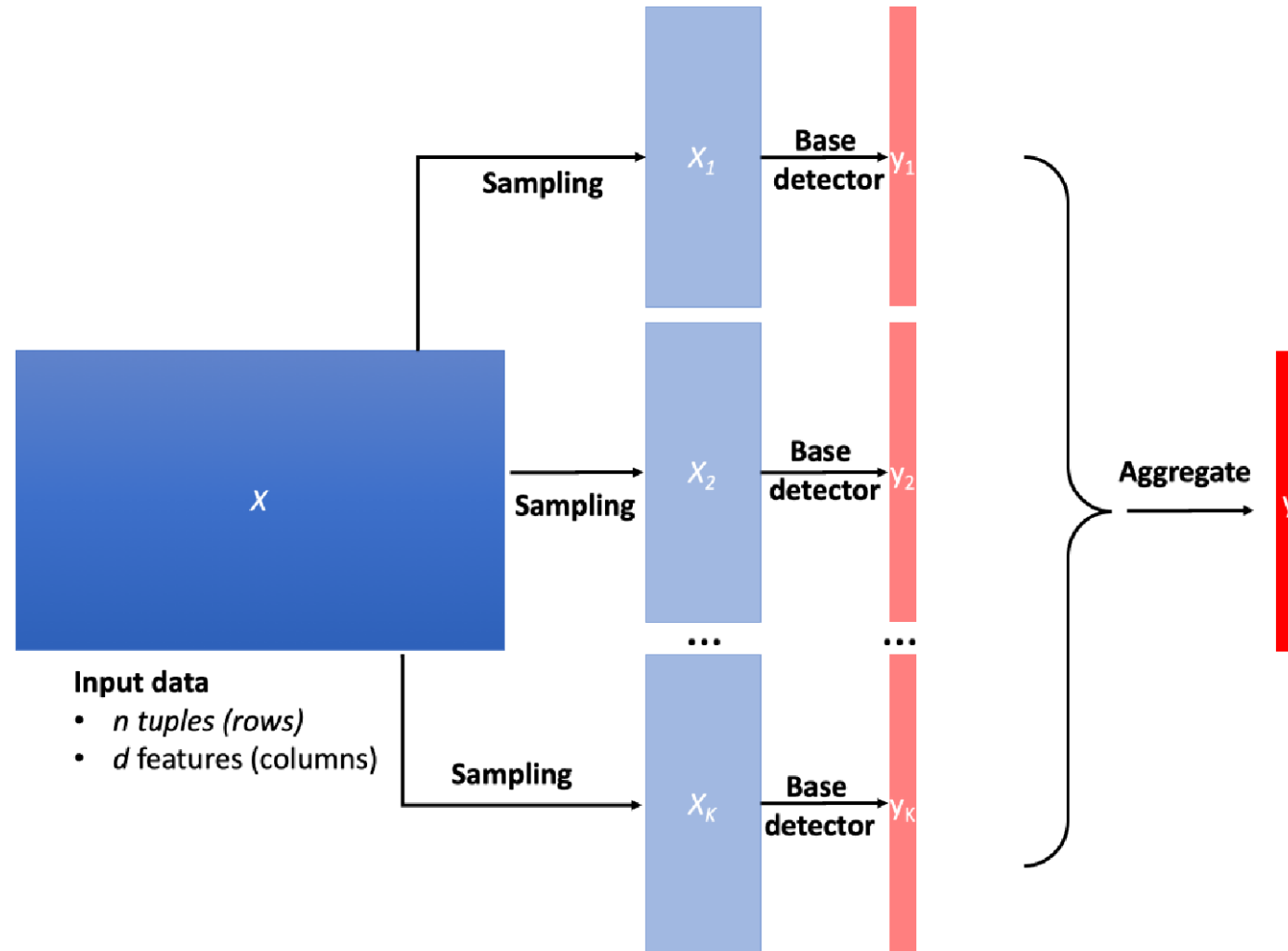


Outlier Detection Ensemble

- ❑ Input: a dataset X with n data tuples (rows) in d -dimensional space (columns)
- ❑ For each base detector, represent the input dataset in a random subspace X_i ($i = 1, \dots, K$) whose dimensionality d' is much smaller than the original feature dimensionality (i.e., $d' \ll d$), and then use an off-the-shelf outlier detection method (e.g., LOF) to assign each data tuple with an outlier-ness score
- ❑ Represent the outlier-ness scores of all n data tuples in the form of a vector $\mathbf{y}_i$ of length n ($i = 1, \dots, K$)
- ❑ Aggregate the detection results from the K base detectors to obtain the overall outlier-ness scores for all input data tuples in the form of another vector $\mathbf{y}$ of length n
- ❑ Flag data tuples with the highest overall outlier-ness scores as outliers

Outlier Detection Ensemble



Key Issues

- ❑ How to represent the input data in a random subspace
 - ❑ Feature bagging: randomly select a few original features to form the representation of the input data in a random subspace
 - ❑ Rotated bagging: first generate a random subspace of dimensionality d' ($d' \ll d$) and then project the input data into this subspace
- ❑ How to aggregate the detection results of base detectors
 - ❑ Mean aggregation: the overall outlier-ness score of a given data tuple is the average of its outlier-ness scores from the K base detectors
 - ❑ Max aggregation: the overall outlier-ness score of a given data tuple is the maximum of its outlier-ness scores from the K base detectors
 - ❑ It is important to normalize (e.g., min-max normalization, z-score normalization) the outlier-ness scores of the base detectors before aggregation, so that the outlier-ness score of a base detector will not overwhelm the scores of other base detectors